authorizationToken MQTT/TLS/JSON

Interface Design Description

Service ID: *"authorizationToken"*

Abstract
This document describes the MQTT protocol with TLS payload security and JSON payload encoding variant of the `authorizationToken` service of the ConsumerAuthorization system.

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `authorizationToken` service interface of the ConsumerAuthorization system over MQTT, which provides authorization tokens: a consumer obtains a token for a specific provider, service and operation; the provider verifies it.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	MQTT	3.1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `authorizationToken` service interface

This document provides the Interface Design Description IDD to the *authorizationToken – Service Description* document [5]. The operations, their semantics and the data models are the same as in the HTTP/TLS/JSON IDD [6]; this document specifies the MQTT binding.

The MQTT interface is offered only when the system runs with `mqtt.api.enabled=true`; it is then registered with the interface template `generic_mqtts` (`generic_mqtt` without TLS), the broker address and port (`mqtt.broker.address`, `mqtt.broker.port`) and the base topic `arrowhead/consumer-authorization/authorization-token/`. In the Arrowhead-test-cloud-java deployment the MQTT interfaces of the core systems are disabled.

3 Service Operations

The provider subscribes to one topic per operation, the base topic followed by the operation name:

Operation	Request topic
`generate`	`arrowhead/consumer-authorization/authorization-token/generate`
`verify`	`arrowhead/consumer-authorization/authorization-token/verify`
`get-public-key`	`arrowhead/consumer-authorization/authorization-token/get-public-key`
`register-encryption-key`	`arrowhead/consumer-authorization/authorization-token/register-encryption-key`
`unregister-encryption-key`	`arrowhead/consumer-authorization/authorization-token/unregister-encryption-key`

A consumer publishes a request envelope (MqttRequestTemplate) to the topic and receives the answer (MqttResponseTemplate) on its own `responseTopic`. The `authentication` field carries the requester's identity according to the provider's authentication policy: `IDENTITY-TOKEN//<token>` (outsourced or internal policy), `SYSTEM//<name>` (declared policy) or the requester's X.509 certificate (certificate policy). The `status` of the response uses the HTTP status code numbers (200, 201, 204, 400, 401, 403, 404, 408, 423, 500, 503); on error the payload is an ErrorMessage.

```
{
  "traceId": "a1b2c3",
  "authentication": "IDENTITY-TOKEN//<token>",
  "responseTopic": "arrowhead/response/TemperatureDisplay/a1b2c3",
  "qosRequirement": 1,
  "params": {},
  "payload": {}
}
```

Listing 1: Request envelope.

```
{
  "status": 200,
  "traceId": "a1b2c3",
  "receiver": "TemperatureDisplay",
  "payload": {}
}
```

Listing 2: Response envelope.

3.1 PUBLISH `arrowhead/consumer-authorization/authorization-token/generate`

Operation: generate
Input: AuthorizationTokenGenerationRequest
Output: AuthorizationTokenGenerationResponse

Generates a token of the requested variant for the requester (as consumer) to the provider's target and operation, if authorized.

```
{
  "tokenVariant": "TIME_LIMITED_TOKEN_AUTH",
  "provider": "TemperatureSensor",
```

```
  "targetType": "SERVICE_DEF",
  "target": "temperatureReading",
  "scope": "read"
}
```

Listing 3: Payload of an generate request.

3.2 PUBLISH arrowhead/consumer-authorization/authorization-token/verify

Operation: verify
Input: String
Output: AuthorizationTokenVerifyResponse

Verifies a token presented to the requester (as provider), counts a use of usage-limited tokens and returns its consumer, target and scope.

```
"<token>"
```

Listing 4: Payload of an verify request.

3.3 PUBLISH arrowhead/consumer-authorization/authorization-token/get-public-key

Operation: get-public-key
Output: String

Returns the public key (Base64 X.509 SubjectPublicKeyInfo) with which providers verify JSON Web Tokens.

3.4 PUBLISH arrowhead/consumer-authorization/authorization-token/register-encryption-key

Operation: register-encryption-key
Input: AuthorizationEncryptionKeyRegistrationRequest
Output: String

Registers or replaces the requester's AES key for encrypting its self-contained tokens. Returns the initialization vector (CBC) or an empty string; 201 for a new key.

```
{
  "key": "0123456789abcdef",
  "algorithm": "AES/ECB/PKCS5Padding"
}
```

Listing 5: Payload of an register-encryption-key request.

3.5 PUBLISH arrowhead/consumer-authorization/authorization-token/unregister-encryption-key

Operation: unregister-encryption-key

Removes the requester's encryption key. Returns 200 if removed, 204 if there was none.

4 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Each *struct* denotes a JSON Object with the named fields; fields that are not needed may be omitted.

4.1 struct MqttRequestTemplate

The request envelope.

Object Field	Value Type	Description
"traceId"	String	Identifier chosen by the requester, returned in the response.
"authentication"	String	Identity of the requester (see Section 3).
"responseTopic"	String	Topic on which the response is published.
"qosRequirement"	Number	MQTT QoS of the response (0, 1 or 2).
"params"	Object<String>	Optional parameters (the query parameters of the HTTP binding).
"payload"	Object	The operation input.

4.2 struct MqttResponseTemplate

The response envelope.

Object Field	Value Type	Description
"status"	Number	Status code.
"traceId"	String	The trace identifier of the request.
"receiver"	String	Name of the requester.
"payload"	Object	The operation output, or an ErrorMessage.

4.3 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

4.4 struct AuthorizationEncryptionKeyRegistrationRequest

Registers the requester's key for encrypting its self-contained tokens.

Object Field	Value Type	Description
"key"	String	AES key of the provider (at least 16 bytes).
"algorithm"	String	Encryption algorithm of the key, e.g. `AES/ECB/PKCS5Padding` or `AES/CBC/PKCS5Padding`.

4.5 struct AuthorizationTokenGenerationRequest

Request of the consumer for a token to one provider's service operation.

Object Field	Value Type	Description
"tokenVariant"	String	Requested token variant, as interface security policy (e.g. `TIME_LIMITED_TOKEN_AUTH`, `RSA_SHA512_JSON_WEB_TOKEN_AUTH`).
"provider"	String	Provider system the token is for.
"targetType"	String	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
"target"	String	Service definition (or event type).
"scope"	String	Operation the token is for.

4.6 struct AuthorizationTokenGenerationResponse

A generated token.

Object Field	Value Type	Description
"tokenType"	AuthorizationTokenType	Token type: `TIME_LIMITED_TOKEN`, `USAGE_LIMITED_TOKEN`, `SELF_CONTAINED_TOKEN` or `TRANSLATION_BRIDGE_TOKEN`.
"targetType"	AuthorizationTargetType	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
"token"	String	The token (opaque string, Base64 or JSON Web Token).
"usageLimit"	Number	Number of allowed uses (usage-limited tokens).
"expiresAt"	DateTime	UTC expiry of time-limited tokens.

4.7 struct AuthorizationTokenVerifyResponse

Result of verifying a token presented to a provider.

Object Field	Value Type	Description
"verified"	Boolean	True when the token is valid.
"consumerCloud"	String	Cloud identifier of the consumer system.
"consumer"	String	Name of the consumer system.
"targetType"	AuthorizationTargetType	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
"target"	String	Target of the policy or token: a service definition or event type.
"scope"	String	Operation the token or policy is limited to; empty for all operations.

4.8 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available.

Type	Description
AuthorizationTargetType	One of `SERVICE_DEF`, `EVENT_TYPE` (JSON String).
AuthorizationTokenType	One of `TIME_LIMITED_TOKEN`, `USAGE_LIMITED_TOKEN`, `SELF_CONTAINED_TOKEN`, `TRANSLATION_BRIDGE_TOKEN` (JSON String).
Boolean	JSON Boolean, `true` or `false`.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

5 Go and Rust implementation variants

5.1 Go implementation

Primary system folder: `Eclipse-Arrowhead-go/ah5-core-go/consumer-authorization`.

The system owns its Go module, launcher and build configuration. Use the per-system `Makefile` and `go.mod`; the containing Go tree is the build source, not a historical Arrowhead copy. Ordinary native roles use independent identities and JSON snapshot state rather than the Java database. Do not assume Java SQL migration, clustering or identical transaction/retry semantics. The registry has its own configuration and optional native AMQP monitor; other roles must be checked against their own registrations.

Listening port. Built-in default 127.0.0.1:8445 (Arrowhead default). The shipped config.json sets "listen": "0.0.0.0:8445". Override with "listen" in the configuration file, SERVER_PORT=N or -server.port N; the port overrides keep the configured host. Native design/route evidence: `Eclipse-Arrowhead-go/ah5-core-go/consumer-authorization/documentation`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

5.2 Rust implementation

Primary system folder: `Eclipse-Arrowhead-rust/ah5-core-rust/consumer-authorization`.

The per-system entry point is built by the `arrowhead-core-rust` crate. Build from the primary Rust workspace with the applicable package/binary selection. Documentation links can resolve into the shared core/support catalogue. Independent process identity and state remain required.

The ordinary native profile uses Tokio/Axum and rustls with peer-identity checks and independent locked JSON state. Network effects are not distributed database transactions. The precise trust, CRL reload and persistence guarantees are those of the native profile, not the Java implementation.

Native design/route evidence: `Eclipse-Arrowhead-rust/ah5-core-rust/documentation/ConsumerAuthorization`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

6 References

[1] A. Banks and R. Gupta, “Mqtt version 3.1.1,” OASIS, Standard, 2014.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “authorizationToken – Service Description (ConsumerAuthorization),” Eclipse Arrowhead, SD 5.3.0, 2026, consumerAuthorization-v5.3.0-authorizationToken-SD.pdf.

[6] ——, “authorizationToken HTTP/TLS/JSON – Interface Design Description (ConsumerAuthorization),” Eclipse Arrowhead, IDD 5.3.0, 2026, consumerAuthorization-v5.3.0-authorizationToken-HTTP-IDD.pdf.

7 Revision History

7.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

7.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	